

Submission Cover Sheet

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A Reflection on this Assignment

This reflective report aims to discuss the production of my project on the “Air Quality” dataset. I am interested in ecology and the environment, so the analysis of this dataset provided worthwhile challenge.

Defining an Analytical Question

Once I had chosen a dataset to use, I did some background research to understand the workings of metal oxide pollution sensors. In 2004, reliable metal oxide sensors were not cheap, and failure in the presence of other gases was possible. Careful calibration is vital. (Fine, 2010). Today, cheaper models are available, and a network of sensors would likely be deployed.

I asked two linked analytical questions:

- Is it possible to predict future pollution levels based on historical data?
- Can this data be used to inform residents of the city about potentially risky times, and when to take precautions or to avoid going outside without a mask?

The second question aims to address the practical application of any possible predictions. I felt that this is the crux of the analysis – there could be a genuine impact from the study. I believe that my questions are relevant for the following reasons.

According to the Welsh Government "air pollution in the UK reduces the life expectancy of every person by an average of 7–8 months, with an associated cost of up to £20 billion each year." (WelshGov, 2025). Proactive public health measures are needed.

Reliable pollution forecasts can be used to inform residents about high-risk periods, enabling them to take preventive action. It may also encourage some to switch to less polluting forms of transport.

I believe that answering the questions posed could allow city planners and residents alike to act in an ethical and positive way to reduce pollution. Schemes such as the ULEZ scheme in London have been shown to reduce traffic pollution. (Asthma and Lung, 2023)

Exploration and Processing of the Data

The dataset is incomplete; it is missing values for most features. This can be attributed to intermittent sensor failure.

The NMHC(GT) sensor failed completely early in the original study; it was missing over 90% of values. Most other features are missing approximately 4%, while three were missing 18%. Running this analysis allowed me to treat these features appropriately.

The failed sensor was disregarded completely, while other missing values were imputed using the K-Nearest Neighbours algorithm. This appeared to be the most appropriate method but was not completely effective. For example, I noted that the imputed Temperature had some unlikely values, but most features were filled in suitably. I feel that this is a useful skill to have learned.

Whether the data is accurate is also a consideration. It may be that sensor drift is a factor in the dataset over the timeframe of the dataset. (Fine, 2010).

I drew graphs to review the shape of the data, and to ascertain correlations between fields. I removed a poorly correlating feature as all other pollutants are strongly correlated. Previously, I would have not been confident to remove a feature like this, which I feel shows growth.

I attempted to use a NARMAX model to predict future pollution levels from the past time series data. I used the only modelling library available, called “sysidentpy”. This proved to be complicated to set up to create a reliable model, and information on the library is highly technical. The results were not particularly useful.

I found that heatmaps of pollutant concentration, plotting the hour of the day against weekday (and against the month) were more effective. Pollution levels are highest from 0730 to 0930, and again from 1630-1930 on weekdays. This effect is increased in the winter months, when more drive to work due to the weather. August is comparatively pollution free.

My python programming has improved this year, in particular my work with the pandas library. I have improved my use of lambda functions. I am writing code in a more “pythonic” way. I was pleased that I was able use resampling and pivot tables in an effective way.

Further Study

It would be interesting to obtain further data from a wider grid of sensors placed at known locations. A wider network of sensors would allow for plotting pollution ‘hotspots’ within a city, further informing urban planning and traffic limitation strategies.

The more sensor readings that are available, the easier I believe it would be to spot false readings, particularly where a sensor is wildly out of calibration with its peers. Placing sensors away from traffic may identify pollution from other sources – such as garden bonfires, construction sites, human waste management or landfill sites.

Further atmospheric data that would be interesting to record would include wind direction and wind speed. This would enable analysis of dispersion and identification of pollutant sources. (UKGov, 2024)

Conclusion

This assignment has improved my understanding of data analysis and the challenges of working with real-world datasets.

I learned how to better handle missing data and apply time series models. I also developed a greater appreciation for the role of data in informing public health and urban planning.

I aim to build on these skills by consolidating my understanding of these modelling techniques and hope to contribute to projects with genuine impact to my organisation.

(870 words)

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